7.6.6.1 Determination of refractive index of material of the prism

The preliminary adjustments of the spectrometer are done. The refractive index of the prism can be determined by measuring the angle of the prism A and the angle of minimum deviation D.

(i) Angle of the prism A

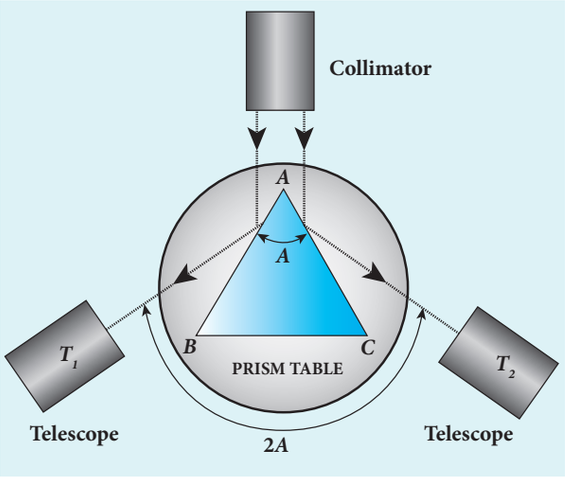


Figure 7.44 Angle of prism

The prism is placed on the prism table with its refracting angle A facing the collimator as shown in Figure 7.44. The slit is illuminated by sodium light (monochromotic light). The parallel rays coming from the collimator fall on the two faces AB and AC and get reflected. The telescope is rotated to the position T_1 and T_2 to capture the reflected rays and the two reading are noted.

The difference between these two readings gives the angle rotated by the telescope, which is twice the angle of the prism. Half of this value gives the angle of the prism A.

(ii) Angle of minimum deviation D

The prism is placed on the prism table so that the light from the collimator falls on a refracting face and the refracted image is observed through the telescope as shown in Figure 7.45. The prism table alone is now rotated so that the angle of deviation decreases. A stage comes when the image stops and returns on further rotation of the prism table. This is ensured by looking through the telescope simultaneously. The reading in this position gives the minimum deviation position.

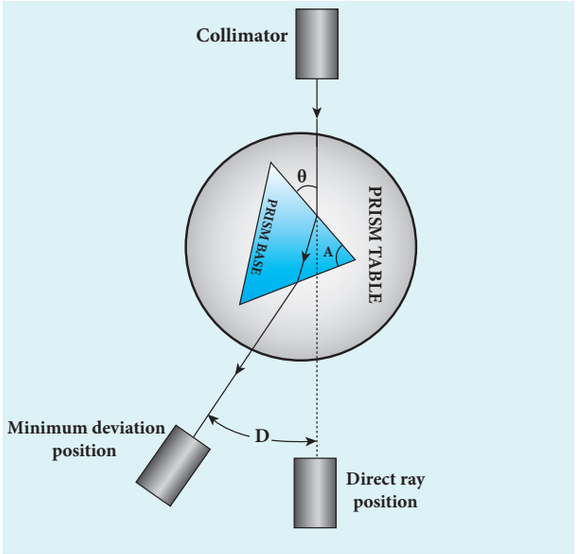


Figure 7.45 Angle of minimum deviation

Now, the prism is removed and the telescope is turned to receive the direct ray and the reading is noted. The difference between the two readings gives the angle of minimum deviation D. The refractive index of the material of the prism n is calculated using the using the equation (6.89),

$$n = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

The refractive index of a liquid may be determined in the same way by using a hollow glass prism filled with the liquid.

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